

Condorcet's paradox for weak preference orderings

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Abstract

We provide in this note some analytical representations for the probability of the Condorcet paradox when voter indifference between alternatives is allowed. The probability of having a Condorcet winner is also considered. Our results show that indifference reduces the likelihood of the paradox, thus improving the ability of the majority rule to aggregate individual preferences. However, indifference also decreases the probability of having a strict Condorcet winner and, for small electorates, the probability of having a Condorcet winner. © 2001 Elsevier Science B.V. All rights reserved.

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1. Introduction

In a collective choice process, a *Condorcet winner* is the alternative that would be able to defeat all other alternatives in a series of majority pairwise comparisons. The fact that a Condorcet winner need not exist constitutes the *Condorcet paradox* (or voting paradox) and gave birth to contemporary social choice theory. Indeed, most of the impossibility theorems that social choice theory has produced can be considered, to some extent, as generalizations of the Condorcet paradox. On a practical level, the voting paradox is troublesome since it sets the problem of the stability of the chosen

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alternative: there will always be another alternative that could defeat the one chosen. Moreover, the presence of the paradox can clearly encourage strategic voting.

Given the theoretical and practical relevance of the voting paradox, a great deal of attention has been paid to the question of its frequency (see Gehrlein, 1997, for an exhaustive survey). Typically, the studies to assess the likelihood of the paradox have assumed that the voters are never indifferent between distinct alternatives, i.e. that the voters are equipped with linear (or strict) preference orders. This is clearly a very strong assumption and one that is not driven by any empirical evidence but rather by a desire to reduce the complexity of the problem. Our purpose in this note is to re-examine the probability of the paradox as well as the probability of Condorcet winners when this assumption is relaxed. To our knowledge, the only attempts to compute the probability of the voting paradox with *weak* (instead of strict) preference orderings are Fishburn and Gehrlein (1980), Jones et al. (1995) and Van Deemen (1999). The results given by Fishburn and Gehrlein are limiting values and are based upon approximations. Van Deemen (1999) provides an exact value but only considers the very specific case with three alternatives and three voters, and the results given by Jones et al. (1995) are estimates obtained from computer simulations. By contrast, we provide in this study some analytical representations and exact values for the paradox probability.

Our analysis also differs from previous work in that we distinguish two types of Condorcet paradoxes. The first one corresponds to the definition used by the above mentioned authors and arises when each alternative is beaten by another one in pairwise majority comparisons: for each alternative A , there exists another alternative B such that the number of voters who prefer B to A is higher than the number of voters who prefer A to B . The second one is a strict version of the first and occurs when the majorities contain more than 50% of the voters; in this case, the Condorcet paradox is said to be *strict*. Note that, when voters' preferences are linear orders, a Condorcet paradox is always strict (at least for an odd number of voters). In addition, the core of the (majority) voting game associated with the collective choice process is empty if and only if a strict Condorcet paradox occurs.

The paper is organized as follows: the framework of our computations is briefly outlined in Section 2. The results we obtain for the strict paradox are presented in Section 3 and Section 4 is devoted to the non-strict version of the paradox. Concluding remarks are given in Section 5.

2. Framework

Let $\{A, B, C, \dots\}$ be a finite set of alternatives (or candidates) and n the number of agents (or voters). Each agent is equipped with a (complete) weak order on the set of alternatives, i.e. each voter is able to completely rank the alternatives, the possibility of ties being admitted. The major part of this paper will focus on the three-alternative case. In this case, the 13 weak orders that voters might have on the alternatives are numbered as follows:

1. ABC 2. ACB 3. BAC 4. BCA 5. CAB 6. CBA 7. $(AB)C$ 8. $(AC)B$
9. $(BC)A$ 10. $A(BC)$ 11. $B(AC)$ 12. $C(AB)$ 13. (ABC)

where ABC means that A is preferred to B and B to C , and $(AB)C$ means that there is an indifference between A and B and both A and B are preferred to C . Let n_i denote the number of voters having the weak order i and p_i the probability that a randomly selected voter will have the associated weak order. A voter *profile*, or simply a *profile*, denotes a specific combination of the n_i 's that sum to n . The two most common assumptions that have been used in the literature to describe the likelihood that various profiles observed are:

- *Impartial Anonymous Culture (IAC)*: Each possible profile is equally likely to occur.
- *Impartial Culture (IC)*: A randomly selected voter is equally likely to have each of the possible preferences on the candidates, that is $p_i = 1/M$, where M denotes the total number of weak orders on the set of candidates ($M = 13$ in the three-alternative case).

Let S denote the set of all possible profiles. The cardinality of S in the three-alternative case is given as:

$$|S| = \frac{(n+1)(n+2)(n+3)\dots(n+12)}{12!}.$$

Consequently, the probability of a given profile with three candidates under the IAC assumption is:

$$\Pr(n_1, n_2, \dots, n_{13}) = \frac{12!}{\prod_{i=1}^{12} (n+i)}.$$

Under the IC assumption, we have the corresponding (multinomial) probability:

$$\Pr(n_1, n_2, \dots, n_{13}) = \frac{n!}{n_1! n_2! \dots n_{13}!} 13^{-n}$$

The results given in Van Deemen (1999), Jones et al. (1995) and Fishburn and Gehrlein (1980) are based upon the IC model. In the present paper, we shall make use of both of the above probabilistic models in order to obtain more robust conclusions. The reader is referred to Berg and Lepelley (1994) for an analysis and a comparison of these models. We shall just mention here that IC assumes that voters act in a completely independent way, whereas IAC implies some degree of dependence between voters preferences.

3. Probability of having an empty core (strict Condorcet's paradox)

Let n_{AB} denote the number of voters who prefer alternative A to alternative B . The core of the majority game is empty if and only if, for every alternative A , there exists an alternative B such that $(n_{BA}/n) > (1/2)$. We call such a situation a strict Condorcet paradox and the current section is (mainly) devoted to the computation of the probability

of this paradox. Let us focus on the three-alternative case. In such a case, a strict Condorcet paradox arises if either $[(n_{AB}/n) > (1/2) \text{ and } (n_{BC}/n) > (1/2) \text{ and } (n_{CA}/n) > (1/2)]$ (cycle of type 1) or $[(n_{BA}/n) > (1/2) \text{ and } (n_{AC}/n) > (1/2) \text{ and } (n_{CB}/n) > (1/2)]$ (cycle of type 2). By the symmetry of our probabilistic models, these two types of cycles have the same probability of occurrence. Assuming n is odd, a cycle of type 1 occurs when:

$$n_1 + n_2 + n_5 + n_8 + n_{10} \geq \frac{n+1}{2},$$

$$n_1 + n_3 + n_4 + n_7 + n_{11} \geq \frac{n+1}{2},$$

$$n_4 + n_5 + n_6 + n_9 + n_{12} \geq \frac{n+1}{2}.$$

Given that $\sum_{i=1}^{13} n_i = n$, it can be established that the above inequalities are equivalent to

$$\left\{ \begin{array}{l} 1 \leq n_1 \leq \frac{n-1}{2} \\ \frac{n+1}{2} - n_1 \leq \alpha \leq \frac{n-1}{2} \\ n_1 + \alpha - \frac{n-1}{2} \leq n_4 \leq \alpha \\ \frac{n+1}{2} - n_1 \leq \beta \leq n - n_1 - \alpha \\ n_1 + \alpha + \beta - n_4 - \frac{n-1}{2} \leq n_5 \leq \beta \\ \text{Max}\left(0, \frac{n+1}{2} - n_4 - n_5\right) \leq \gamma \leq n - n_1 - \alpha - \beta \\ 0 \leq n_3 \leq \alpha - n_4 \\ 0 \leq n_7 \leq \alpha - n_3 - n_4 \\ 0 \leq n_2 \leq \beta - n_5 \\ 0 \leq n_8 \leq \beta - n_2 - n_5 \\ 0 \leq n_6 \leq \gamma \\ 0 \leq n_9 \leq \gamma - n_6 \end{array} \right. \quad (1)$$

with α, β, γ defined as follows:

$$\alpha = n_3 + n_4 + n_7 + n_{11}$$

$$\beta = n_2 + n_5 + n_8 + n_{10}$$

$$\gamma = n_6 + n_9 + n_{12}$$

and

$$n_{13} = n - \alpha - \beta - \gamma - n_1.$$

These relations make possible the calculation of the desired probabilities by computer enumeration for small values of n . Under the IC assumption, the probability of having a strict Condorcet paradox is given as

$$2 \sum \frac{n! 13^{-n}}{n_1! n_2! \dots n_9! (\beta - n_2 - n_5 - n_8)! (\alpha - n_3 - n_4 - n_7)! (\gamma - n_6 - n_9)! (n - \alpha - \beta - \gamma - n_1)!}$$

and the corresponding probability under the IAC assumption is

$$2 \sum \frac{12!}{\prod_{i=1}^{12} (n+i)} = 2 \frac{12!}{\prod_{i=1}^{12} (n+i)} \sum 1$$

where $\sum^1$ is a twelve-summation function with summation indices consistent with (1).

Moreover, it is possible to obtain a closed formula under the IAC assumption by removing the Max argument from (1).

Proposition 1. *For n odd and $n+1$ multiple of four, the probability of having a strict Condorcet paradox in three-alternative elections under the IAC assumption is given by*

$$\frac{(n+13)(n+17)(n+21)(n^6 + 66n^5 + 1603n^4 + 17,292n^3 + 75,967n^2 + 63,426n - 156,915)}{4096(n+12)(n+11)(n+10)(n+8)(n+7)(n+6)(n+4)(n+3)(n+2)}$$

Proof. See Appendix A. □

From this relation, we obtain that, when n tends to infinity, the probability of a strict Condorcet paradox tends to $1/4096 = 0.00024$.

Such a closed form relation cannot be obtained under the IC assumption. The limiting probability of having an empty core is, however, easy to compute under IC: appealing to the Central Limit theorem, it is straightforward to prove the following.

Proposition 2. *Suppose n tends to infinity. Then, the probability of having a strict Condorcet paradox is zero under the IC assumption.*

Proof. The occurrence of a strict Condorcet paradox implies that inequalities of the following form must hold:

$$n_{AB} > \frac{n+1}{2}.$$

Observing that

- (i) the proportion of weak orders with A preferred to B is less than 50%,
- (ii) n_{AB} is normally distributed under IC when n becomes large,

Table 1
Probability of having a strict Condorcet paradox for three candidates and n voters

n	IAC assumption		IC assumption	
	weak orders	linear orders	weak orders	linear orders
3	0.00440	0.03571	0.00546	0.05556
5	0.00388	0.04762	0.00339	0.06944
7	0.00314	0.05303	0.00192	0.07502
9	0.00256	0.05594	0.00109	0.07798
11	0.000214	0.05769	0.00062	0.07981
13	0.00182	0.05882	0.00036	0.08107
15	0.00159	0.05960	0.00021	0.08198
17	0.00141	0.06015	0.00012	0.08267
19	0.00127	0.06056	0.00007	0.08322
21	0.00115	0.06087	0.00004	0.08365
⋮	⋮	⋮	⋮	⋮
∞	0.00024	0.0625	0	0.08774

it is easy to conclude that the probability of having $n_{AB} > ((n + 1)/2)$ tends to zero when n tends to infinity. □

Observe that this result holds for *any* number of candidates (not only in three-alternative elections).

Computed values of the probability of having a strict Condorcet paradox with weak orders are shown in Table 1 (the corresponding values for linear orders are taken from Gehrlein, 1997). It turns out that *the possibility of indifference considerably decreases the probability of having an empty core in three-alternative elections.*¹ The contrast between the case with linear orders and the case with weak orders is particularly sharp when an increasing number of candidates are considered: for a large number of candidates, the probability of having an empty core tends to unity with linear orders (see, e.g., Gehrlein, 1997) whereas this probability tends to zero when indifference is allowed, and this conclusion holds under IAC as well as under IC since these two probabilistic models tend to coincide when the number of alternatives increases.

It is worth noting, however, that the absence of a strict Condorcet paradox does not mean that there exists a Condorcet winner when indifference is allowed. In order to make precise this observation, we have computed the probability of having what we call a strict Condorcet winner, i.e. a candidate preferred to every other candidate by majorities of more than 50% of the voters. In three-alternative elections, an alternative A is the strict Condorcet winner if and only if:

$$n_1 + n_2 + n_5 + n_8 + n_{10} \geq \frac{n + 1}{2}, \quad \text{and} \quad n_1 + n_2 + n_3 + n_7 + n_{10} \geq \frac{n + 1}{2}, \quad (2)$$

with n odd. Starting from these inequalities, we obtain the following proposition.

¹ The value we obtain under IC for $n = 3$ corroborates Van Deemen's result (Van Deemen, 1999).

Proposition 3. For n odd, the probability of having a strict Condorcet winner in three-alternative elections under the IAC assumption is given by

$$\frac{3(37n^6 + 1968n^5 + 41,191n^4 + 429,864n^3 + 2,341,099n^2 + 6,337,896n + 6,814,665)}{512(n+12)(n+10)(n+8)(n+6)(n+4)(n+2)}$$

Suppose now that n tends to infinity and that the IC assumption prevails. Then, the probability of having a strict Condorcet winner is zero.

Proof. As $\sum_{i=1}^{13} n_i = n$, inequalities (2) are equivalent to:

$$\left\{ \begin{array}{l} 0 \leq \alpha \leq \frac{n-1}{2} \\ 0 \leq \beta \leq \frac{n-1}{2} - \alpha \\ 0 \leq \gamma \leq \frac{n-1}{2} - \alpha \\ 0 \leq n_4 \leq \alpha \\ 0 \leq n_6 \leq \alpha - n_4 \\ 0 \leq n_9 \leq \alpha - n_4 - n_6 \\ 0 \leq n_{11} \leq \alpha - n_4 - n_6 - n_9 \\ 0 \leq n_{12} \leq \alpha - n_4 - n_6 - n_9 - n_{11} \\ 0 \leq n_1 \leq n - \alpha - \beta - \gamma \\ 0 \leq n_2 \leq n - \alpha - \beta - \gamma - n_1 \\ 0 \leq n_5 \leq \beta \\ 0 \leq n_3 \leq \gamma \end{array} \right. \quad (3)$$

with $\alpha = n_4 + n_6 + n_9 + n_{11} + n_{12} + n_{13}$, $\beta = n_5 + n_8$ and $\gamma = n_3 + n_7$. The number of situations in which A is a strict Condorcet winner is thus given as

$$\begin{aligned} & \sum_{\alpha=0}^{(n-1)/2} \sum_{\beta=0}^{(n-1)/2-\alpha} \sum_{\gamma=0}^{(n-1)/2-\alpha-\beta} \sum_{n_4=0}^{\alpha} \sum_{n_6=0}^{\alpha-n_4} \sum_{n_9=0}^{\alpha-n_4-n_6} \sum_{n_{11}=0}^{\alpha-n_4-n_6-n_9} \\ & \times \sum_{n_{12}=0}^{\alpha-n_4-n_6-n_9-n_{11}} \sum_{n_1=0}^{n-\alpha-\beta-\gamma-n_{12}} \sum_{n_2=0}^{n-\alpha-\beta-\gamma-n_1} \sum_{n_5=0}^{\beta} \sum_{n_3=0}^{\gamma} 1. \end{aligned}$$

This multiple sum can be reduced by using the formulas for sums of powers of integers and the desired probability is obtained by multiplying by three and dividing by $|S|$. The second assertion in the proposition follows from the same argument as the one used in Proposition 2. $\square$

Computed values of the probability of having a strict Condorcet winner in three-alternative elections are given in Table 2. Comparison with what is obtained for linear orders shows that, as expected, allowing indifference completely changes the picture (espe-

Table 2

Probability of having a strict Condorcet winner for three candidates and n voters

n	IAC assumption		IC assumption	
	weak orders	linear orders	weak orders	linear orders
3	0.54066	0.96429	0.46563	0.94444
5	0.46736	0.95238	0.31113	0.93056
7	0.42254	0.94697	0.20191	0.92498
9	0.39184	0.94406	0.12773	0.92202
11	0.36932	0.94231	0.07915	0.92019
13	0.35204	0.94118	0.04823	0.91893
15	0.33833	0.94040	0.02900	0.91802
17	0.32718	0.93985	0.01716	0.91733
19	0.31791	0.93944	0.00994	0.91678
21	0.31009	0.93913	0.00569	0.91635
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
∞	0.21680	0.93750	0	0.91226

cially when IC is assumed) and *makes the existence of a strict Condorcet winner a rather rare event*. It is of interest to point out that under IC, the probability for an alternative to be dominated, that is to say to be beaten by more than 50% of the voters in some pairwise comparison, tends to zero in the limit of the voters. Since the core of a voting game is defined as the set of undominated alternatives, it follows that the probability that all the alternatives belong to the core tends to one when the number of voters is large.

4. Condorcet's paradox probability

A Condorcet paradox occurs when, for each alternative A , there exists an alternative B such that $n_{BA} > n_{AB}$. The existence of a Condorcet paradox means that there is no Condorcet winner (a Condorcet winner being defined as a candidate—say A —such that $n_{AB} > n_{BA}$ for every other candidate B) but, when indifference is allowed, the converse is not true: even for odd values of the number of voters, there exist some profiles with no Condorcet winner and no Condorcet paradox (one can have, for instance, $n_{BA} = n_{AB}$). It is clear, however, that such profiles become rare when the number of alternatives increases: for large electorates, the probability of Condorcet paradox tends to be equal to the probability of having no Condorcet winner.

We have not been able to derive a closed formula for the desired probabilities under the IAC assumption. It is possible, however, to compute the limiting probability of having a Condorcet paradox under the IC assumption for the cases of three alternatives and four alternatives.²

² The figures given in Proposition 4 are computed as special forms of orthant probabilities of a multivariate normal distribution (bivariate for the three-alternative case and trivariate for the four-alternative case). Closed form representations for these orthant probabilities exist only for bivariate and trivariate distributions.

Proposition 4. Suppose n tends to infinity. Then, the probability of having a Condorcet paradox under the IC assumption is 5.35% in three-alternative elections and 12.08% in four-alternative elections.³

Proof. Consider first the three-alternative case. We shall deduce the desired probability from the probability of having a Condorcet winner. Let $\Pr_A(3, \infty, \text{IC})$ be the probability that A is the Condorcet winner under IC when n is large. Alternative A is a Condorcet winner if and only if:

$$n_1 + n_2 + n_5 + n_8 + n_{10} > n_3 + n_4 + n_6 + n_9 + n_{11},$$

$$n_1 + n_2 + n_3 + n_7 + n_{10} > n_4 + n_5 + n_6 + n_9 + n_{12}.$$

Let X_1 and X_2 define two discrete variables (for each voter) with associated probabilities given by:

$$X_1 = +1: \quad p_1 + p_2 + p_5 + p_8 + p_{10}$$

$$-1: \quad p_3 + p_4 + p_6 + p_9 + p_{11}$$

$$0: \quad \text{otherwise}$$

$$X_2 = +1: \quad p_1 + p_2 + p_3 + p_7 + p_{10}$$

$$-1: \quad p_4 + p_5 + p_6 + p_9 + p_{12}$$

$$0: \quad \text{otherwise}$$

In words, for a given voter j , X_1 is equal to $+1$ if j strictly refers A to B (the probability of this event is $p_1 + p_2 + p_5 + p_8 + p_{10}$), X_1 is equal to -1 if j strictly prefers B to A (probability: $p_3 + p_4 + p_6 + p_9 + p_{11}$) and to 0 if j is indifferent between A and B . Similarly, the value of X_2 is based upon j 's preference between A and C . Let $\bar{X}_i$ denote the mean value of X_i over the n voters. If $\bar{X}_1 > 0$ and $\bar{X}_2 > 0$ then A will be the Condorcet winner. Given that $p_i = 1/13 \quad \forall i$ under IC, it can be checked that $\Pr_A(3, \infty, \text{IC})$ is the positive orthant probability that $\bar{X}_1\sqrt{n} \geq E(\bar{X}_1\sqrt{n})$ and $\bar{X}_2\sqrt{n} \geq E(\bar{X}_2\sqrt{n})$. By appealing to the Central Limit Theorem, this distribution is bivariate normal and the Sheppard's Theorem of Median Dichotomy (Kendall and Stuart, 1963) can be applied to obtain:

$$\Pr_A(3, \infty, \text{IC}) = \frac{1}{4} + \frac{1}{2\pi} \arcsin(\rho_{12}).$$

Thus, we have to calculate the correlation coefficient ρ_{12} . Since the expected values of X_1 and X_2 are zero, the variances of X_1 and X_2 are equal to $10/13$ and the covariance is equal to $4/13$. Therefore, $\rho_{12} = (2/5)$ and $\Pr_A(3, \infty, \text{IC}) = 0.31549$. By the symmetry of IC, each candidate is equally likely to be the Condorcet winner and the probability of having a Condorcet winner is $3\Pr_A(3, \infty, \text{IC}) = 0.946484$. Therefore, the probability of having a Condorcet paradox is $1 - 0.946484 = 0.053516$. Consider now the case with four alternatives. We can define three discrete variables Z_1 , Z_2 and Z_3 in such a way

³ The results we obtain are consistent with Fishburn and Gehrlein (1980) and Jones et al. (1995).

Table 3
Probability of having a Condorcet paradox for three candidates and n voters

n	IAC assumption weak orders	IC assumption linear orders
3	0.00440	0.00546
5	0.00711	0.00663
7	0.01004	0.00926
9	0.01281	0.01196
11	0.01528	0.01434
13	0.01743	0.01638
15	0.02000	0.01829
$\vdots$	$\vdots$	$\vdots$
∞	0.042 ^e	0.05352

e: estimate.

that alternative A will be the Condorcet winner if and only if $\bar{Z}_1 > 0$, $\bar{Z}_2 > 0$ and $\bar{Z}_3 > 0$ (this easy but tedious exercise is left to the reader). From the Sheppard's Theorem of Median Dichotomy, we then obtain that the probability $\Pr_A(4,\infty,IC)$ of having A as a Condorcet winner is given as:

$$\Pr_A(4,\infty,IC) = \frac{1}{8} + \left(\frac{1}{4\pi} (\arcsin(\rho_{12}) + \arcsin(\rho_{13}) + \arcsin(\rho_{23})) \right).$$

with ρ_{12} , ρ_{13} and ρ_{23} equal to $12/31$. Thus, we obtain $\Pr_A(4,\infty,IC) = 0.21989$ and the probability of having a Condorcet paradox is $1 - 4\Pr_A(4,\infty,IC)$. $\square$

We have also resorted to computer enumeration in order to obtain exact probabilities for small values of n (under IC as well as under IAC).⁴ The results we have obtained for three-candidate elections are shown in Tables 3 and 4 (the corresponding probabilities for the case with linear orders are the same as in Tables 1 and 2, since in this case a Condorcet paradox occurs if and only if a strict Condorcet paradox occurs).

It appears from Table 3 that the probability of having a Condorcet paradox for weak orders is quite small, especially for a small number of voters: we conclude once again that *indifference tends to decrease the likelihood that a paradox occurs*. Table 4 leads to the following rather unexpected conclusion: when replacing linear orders by weak orders, the probability of having a Condorcet winner *decreases for small electorates* (such as committees) *but increases for large electorates*; and this increase is still observable for four candidates: in four-alternative elections, the limiting probability

⁴ The probability of having a Condorcet paradox under IC for a given n is computed by evaluating the following relation

$$2 \sum_{(n_1, \dots, n_{13}) \in S} \frac{\delta n!}{n_1! \dots n_{13}!} 13^{-n}$$

where $\delta = 1$ if $(n_{AB} > n_{BA}, n_{BC} > n_{CB} \text{ and } n_{CA} > n_{AC})$ and $\delta = 0$ otherwise. A similar approach is used to compute the other probabilities for $n \in \{3, 5, \dots, 15\}$.

Table 4

Probability of having a Condorcet winner for three candidates and n voters

n	IAC assumption weak orders	IC assumption weak orders
3	0.71209	0.66363
5	0.74806	0.69178
7	0.77751	0.71953
9	0.80030	0.74145
11	0.81496	0.75865
13	0.83240	0.76855
15	0.87453	0.77369
⋮	⋮	⋮
∞	0.958 ^e	0.94649

e: estimate.

under IC is 74.87% with linear orders (see Gehrlein, 1997) and 87.96% with weak orders.

5. Concluding comments

We have analyzed in this note the effect of introducing indifference in individual preferences on the likelihood of Condorcet's paradox. For the two versions of the paradox we have considered, we obtain that its occurrence is substantially rarer when preferences are represented by weak orders rather than by linear orders, and this conclusion holds for both of the probabilistic assumptions we have used. In this sense, it can be said that, *when indifference is allowed, then there is an improvement in the ability of Condorcet-extension methods to aggregate individual preferences*. Interestingly, a similar conclusion has recently been obtained—independently—by Gehrlein and Valognes (2001) in a framework slightly different from ours, and by Crès (1999) in a completely different setting. These results suggest that the presence of indifference could constitute a possible answer to the famous question posed by Tullock (1981) “Why so much stability?”.

Our findings, however, give a more nuanced picture than the previous work on the impact of indifference. Two observations are in order. Firstly, the distinction between the strict version of the paradox and the nonstrict version appears to be relevant: when the number of agents is large, the probability of having a strict Condorcet paradox (an empty core) tends to zero, whereas the frequency of a Condorcet paradox is small but remains nonnegligible. From this point of view, the result provided by Van Deemen (1999) is rather misleading because in the particular case studied by him (three agents, three alternatives), only a strict version of the paradox can occur.

Secondly, it is important to emphasize that less instability (less cycles) does not necessarily mean more stability (existence of a Condorcet winner) when indifference is allowed. In particular, the fact that indifference warrants a nonempty core for large

electorates does not facilitate the collective choice process because indifference also implies that all the alternatives will belong to the core.

Appendix A

Let Y be the set of voting situations $(n_1, \dots, n_{13})$ satisfying the set of inequalities (1). In order to remove the Max argument from the limit of γ , we partition Y into 13 subsets defined in the following way (in each of these subsets, the limits of n_3 , n_7 , n_2 , n_8 , n_6 , n_9 are the same as in (1) and $n+1$ is assumed to be a multiple of four):

$$Y_1: 1 \leq n_1 \leq \frac{n+1}{4}, \frac{n+1}{2} - n_1 \leq \alpha \leq \frac{n-1}{2},$$

$$n_1 + \alpha - \frac{n-1}{2} \leq n_4 \leq n_1, \frac{n+1}{2} - n_1 \leq \beta \leq \frac{n+1}{2} - n_4,$$

$$n_1 + \alpha + \beta - n_4 - \frac{n-1}{2} \leq n_5 \leq \beta, \frac{n+1}{2} - n_4 - n_5 \leq \gamma \leq n - n_1 - \alpha - \beta.$$

$$Y_2: 1 \leq n_1 \leq \frac{n+1}{4}, \frac{n+1}{2} - n_1 \leq \alpha \leq \frac{n-1}{2},$$

$$n_1 + \alpha - \frac{n-1}{2} + 1 \leq n_4 \leq n_1, \frac{n+1}{2} - n_4 + 1 \leq \beta \leq n - n_1 - \alpha,$$

$$n_1 + \alpha + \beta - n_4 - \frac{n-1}{2} \leq n_5 \leq \frac{n+1}{2} - n_4,$$

$$\frac{n+1}{2} - n_4 - n_5 \leq \gamma \leq n - n_1 - \alpha - \beta.$$

$$Y_3: 1 \leq n_1 \leq \frac{n+1}{4}, \frac{n+1}{2} - n_1 \leq \alpha \leq \frac{n-1}{2},$$

$$n_1 + \alpha - \frac{n-1}{2} + 1 \leq n_4 \leq n_1, \frac{n+1}{2} - n_4 + 1 \leq \beta \leq n - n_1 - \alpha,$$

$$\frac{n+1}{2} - n_4 + 1 \leq n_5 \leq \beta, 0 \leq \gamma \leq n - n_1 - \alpha - \beta.$$

$$Y_4: 1 \leq n_1 \leq \frac{n+1}{4}, \frac{n+1}{2} - n_1 \leq \alpha \leq \frac{n-1}{2},$$

$$n_1 + 1 \leq n_4 \leq \alpha, \frac{n+1}{2} - n_1 \leq \beta \leq n - n_1 - \alpha,$$

$$n_1 + \alpha + \beta - n_4 - \frac{n-1}{2} \leq n_5 \leq \frac{n+1}{2} - n_4,$$

$$\frac{n+1}{2} - n_4 - n_5 \leq \gamma \leq n - n_1 - \alpha - \beta.$$

$$Y_5: 1 \leq n_1 \leq \frac{n+1}{4}, \frac{n+1}{2} - n_1 \leq \alpha \leq \frac{n-1}{2},$$

$$n_1 + 1 \leq n_4 \leq \alpha, \frac{n+1}{2} - n_1 \leq \beta \leq n - n_1 - \alpha,$$

$$\frac{n+1}{2} - n_4 + 1 \leq n_5 \leq \beta, 0 \leq \gamma \leq n - n_1 - \alpha - \beta.$$

$$Y_6: \frac{n+1}{4} + 1 \leq n_1 \leq \frac{n-1}{2}, \frac{n+1}{2} - n_1 \leq \alpha \leq n_1,$$

$$n_1 + \alpha - \frac{n-1}{2} \leq n_4 \leq \alpha, \frac{n+1}{2} - n_1 \leq \beta \leq \frac{n+1}{2} - n_4,$$

$$n_1 + \alpha + \beta - n_4 - \frac{n-1}{2} \leq n_5 \leq \beta,$$

$$\frac{n+1}{2} - n_4 - n_5 \leq \gamma \leq n - n_1 - \alpha - \beta.$$

$$Y_7: \frac{n+1}{4} + 1 \leq n_1 \leq \frac{n-1}{2}, \frac{n+1}{2} - n_1 \leq \alpha \leq n_1,$$

$$n_1 + \alpha - \frac{n-1}{2} + 1 \leq n_4 \leq n_1, \frac{n+1}{2} - n_4 + 1 \leq \beta \leq n - n_1 - \alpha,$$

$$n_1 + \alpha + \beta - n_4 - \frac{n-1}{2} \leq n_5 \leq \frac{n+1}{2},$$

$$\frac{n+1}{2} - n_4 - n_5 \leq \gamma \leq n - n_1 - \alpha - \beta.$$

$$Y_8: \frac{n+1}{4} + 1 \leq n_1 \leq \frac{n-1}{2}, \frac{n+1}{2} - n_1 \leq \alpha \leq n_1,$$

$$n_1 + \alpha - \frac{n-1}{2} + 1 \leq n_4 \leq n_1, \frac{n+1}{2} - n_4 + 1 \leq \beta \leq n - n_1 - \alpha,$$

$$\frac{n+1}{2} - n_4 + 1 \leq n_5 \leq \beta, 0 \leq \gamma \leq n - n_1 - \alpha - \beta.$$

$$Y_9: \frac{n+1}{4} + 1 \leq n_1 \leq \frac{n-1}{2}, n_1 + 1 \leq \alpha \leq \frac{n-1}{2},$$

$$n_1 + \alpha - \frac{n-1}{2} \leq n_4 \leq n_1, \frac{n+1}{2} - n_1 \leq \beta \leq \frac{n+1}{2} - n_4,$$

$$n_1 + \alpha + \beta - n_4 - \frac{n-1}{2} \leq n_5 \leq \beta,$$

$$\frac{n+1}{2} - n_4 - n_5 \leq \gamma \leq n - n_1 - \alpha - \beta.$$

$$Y_{10}: \frac{n+1}{4} + 1 \leq n_1 \leq \frac{n-1}{2}, n_1 + 1 \leq \alpha \leq \frac{n-1}{2},$$

$$n_1 + \alpha - \frac{n-1}{2} + 1 \leq n_4 \leq n_1, \frac{n+1}{2} - n_4 + 1 \leq \beta \leq n - n_1 - \alpha,$$

$$n_1 + \alpha + \beta - n_4 - \frac{n-1}{2} \leq n_5 \leq \frac{n+1}{2} - n_4,$$

$$\frac{n+1}{2} - n_4 - n_5 \leq \gamma \leq n - n_1 - \alpha - \beta.$$

$$Y_{11}: \frac{n+1}{4} + 1 \leq n_1 \leq \frac{n-1}{2}, n_1 + 1 \leq \alpha \leq \frac{n-1}{2},$$

$$n_1 + \alpha - \frac{n-1}{2} + 1 \leq n_4 \leq n_1, \frac{n+1}{2} - n_4 + 1 \leq \beta \leq n - n_1 - \alpha,$$

$$\frac{n+1}{2} - n_4 + 1 \leq n_5 \leq \beta, 0 \leq \gamma \leq n - n_1 - \alpha - \beta.$$

$$Y_{12}: \frac{n+1}{4} + 1 \leq n_1 \leq \frac{n-1}{2}, n_1 + 1 \leq \alpha \leq \frac{n-1}{2},$$

$$n_1 + 1 \leq n_4 \leq \alpha, \frac{n+1}{2} - n_1 \leq \beta \leq n - n_1 - \alpha,$$

$$n_1 + \alpha + \beta - n_4 - \frac{n-1}{2} \leq n_5 \leq \frac{n+1}{2} - n_4,$$

$$\frac{n+1}{2} - n_4 - n_5 \leq \gamma \leq n - n_1 - \alpha - \beta.$$

$$Y_{13}: \frac{n+1}{4} + 1 \leq n_1 \leq \frac{n-1}{2}, n_1 + 1 \leq \alpha \leq \frac{n-1}{2},$$

$$n_1 + 1 \leq n_4 \leq \alpha, \frac{n+1}{2} - n_1 \leq \beta \leq n - n_1 - \alpha,$$

$$\frac{n+1}{2} - n_4 + 1 \leq n_5 \leq \beta, 0 \leq \gamma \leq n - n_1 - \alpha - \beta.$$

For all i belonging to $\{1, \dots, 13\}$, the cardinality of Y_i can be computed by sequential summation using the formulas for sums of powers of integers. We obtain for Y_1 :

$$\begin{aligned} |Y_1| = & \sum_{n_1=1}^{(n+1)/4} \sum_{\alpha=(n+1)/(2-n_1)}^{(n-1)/2} \sum_{n_4=n_1+\alpha-(n-1)/2}^{n_1} \sum_{\beta=(n+1)/(2-n_1)}^{(n+1)/(2-n_4)} \\ & \times \sum_{n_5=n_1+\alpha+\beta-n_4-(n-1)/2}^{\beta} \sum_{\gamma=(n+1)/(2-n_4-n_5)}^{n-n_1-\alpha-\beta} \sum_{n_7=0}^{\alpha-n_4\alpha-n_3-n_4\beta-n_5} \sum_{n_2=0} \end{aligned}$$

$$\begin{aligned}
& \times \sum_{n_8=0}^{\beta-n_2-n_5} \sum_{n_6=0}^{\gamma} \sum_{n_9=0}^{\gamma-n_6} 1 = \{(n+29)(n+25)(n+21)(n+17)(n+13) \\
& \times (n+9)(n+5)(n+1)(277n^4 + 8604m^3 + 92,486n^2 + 346,764n \\
& + 913,869)\} / 8,036,313,307,545,600
\end{aligned}$$

The cardinality of each of $Y_2, \dots, Y_{13}$ can be computed in a similar way, and we finally obtain the cardinality of Y as:

$$\begin{aligned}
|Y| &= \sum_{i=1}^{13} |Y_i| = \{(n+1)(n+5)(n+9)(n+13)(n+17)(n+21) \\
& \times (n^6 + 66n^5 + 1603n^4 + 17,292n^3 + 75,967n^2 \\
& + 63,426n - 156,915)\} / 3,923,981,107,200
\end{aligned}$$

The probability of a strict Condorcet paradox is then given by $2|Y|/|S|$.

References

- Berg, S., Lepelley, D., 1994. Probability models in voting theory. *Statistica Neerlandica* 48, 133–146.
- Crès, H., 1999. Aggregating coarse individual preferences. Mimeograph. H.E.C. Jouy-en-Josas.
- Fishburn, P.C., Gehrlein, W.V., 1980. The paradox of voting: effects of individual indifference and intransitivity. *Journal of Public Economics* 14, 83–94.
- Gehrlein, W.V., 1997. Condorcet paradox and the Condorcet efficiency of voting rules. *Mathematica Japonica* 40, 173–199.
- Gehrlein, W.V., Valognes, F., 2001. Condorcet efficiency: a preference for indifference. *Social Choice and Welfare* 18, 193–206.
- Jones, B., Radcliff, B., Taber, C., Timponi, R., 1995. Condorcet winners and the paradox of voting: probability calculations for weak preference orders. *American Political Science Review* 89, 137–144.
- Kendall, M.G., Stuart, A., 1963. *The Advanced Theory of Statistics*. Griffin, London.
- Tullock, G., 1981. Why so much stability? *Public Choice* 37, 189–202.
- Van Deemen, A., 1999. The probability of the paradox of voting for weak preference orderings. *Social Choice and Welfare* 16, 171–182.